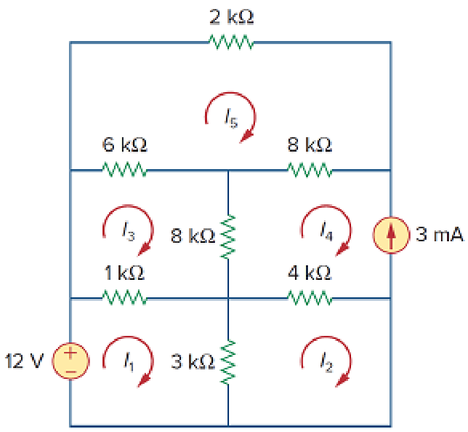
Problem

Find the mesh currents in the circuit of Fig. 3.98 using *MATLAB*.

Figure 3.98:



Step-by-step solution

Step 1 of 6

Consider the following circuit diagram:

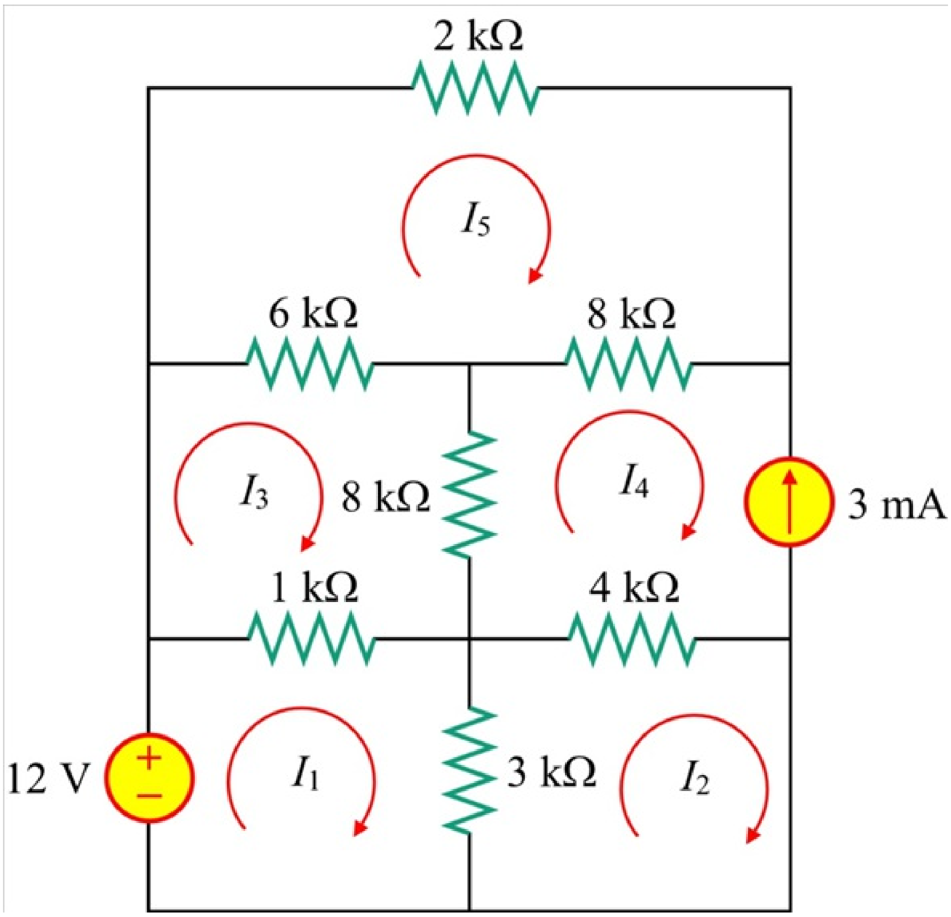


Figure 1

Step 2 of 6

From the Figure 1, the current in the mesh 4 is 3 mA.

$I_4 = -3\text{ mA}$ …… (1)

Apply Kirchhoff's voltage law to the loop 1.

$$(1\text{ k})(I_1 - I_3) + (3\text{ k})(I_1 - I_2) = 12$$
$$(1\text{ k} + 3\text{ k})I_1 - (1\text{ k})I_3 - (3\text{ k})I_2 = 12$$
$$(4\text{ k})I_1 - (3\text{ k})I_2 - (1\text{ k})I_3 = 12$$

$4I_1 - 3I_2 - I_3 = 12\text{ mA}$ …… (2)

Step 3 of 6

Apply Kirchhoff's voltage law to the loop 2.

$$(3\text{ k})(I_2 - I_1) + (4\text{ k})(I_2 - I_4) = 0$$
$$-(3\text{ k})I_1 + (3\text{ k} + 4\text{ k})I_2 - (4\text{ k})I_4 = 0$$
$$-(3\text{ k})I_1 + (7\text{ k})I_2 - (4\text{ k})I_4 = 0$$

$3I_1 - 7I_2 + 4I_4 = 0$ …… (3)

Apply Kirchhoff's voltage law to the loop 3.

$$(1\text{ k})(I_3 - I_1) + (8\text{ k})(I_3 - I_4) + (6\text{ k})(I_3 - I_5) = 0$$
$$-(1\text{ k})I_1 + (1\text{ k} + 8\text{ k} + 6\text{ k})I_3 - (8\text{ k})I_4 - (6\text{ k})I_5 = 0$$
$$-(1\text{ k})I_1 + (15\text{ k})I_3 - (8\text{ k})I_4 - (6\text{ k})I_5 = 0$$

$I_1 - 15I_3 + 8I_4 + 6I_5 = 0$ …… (4)

Step 4 of 6

Apply Kirchhoff's voltage law to the loop 5.

$$(6\text{ k})(I_5 - I_3) + (8\text{ k})(I_5 - I_4) + (2\text{ k})(I_5) = 0$$
$$-(6\text{ k})I_3 - (8\text{ k})I_4 + (6\text{ k} + 8\text{ k} + 2\text{ k})I_5 = 0$$
$$-(6\text{ k})I_3 - (8\text{ k})I_4 + (16\text{ k})I_5 = 0$$

$3I_3 + 4I_4 - 8I_5 = 0$ …… (5)

Write equations (1), (2), (3), (4), and (5) in the matrix form:

$$\begin{bmatrix} 0 & 0 & 0 & 1 & 0 \\ 4 & -3 & -1 & 0 & 0 \\ 3 & -7 & 0 & 4 & 0 \\ 1 & 0 & -15 & 8 & 6 \\ 0 & 0 & 3 & 4 & -8 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \\ I_4 \\ I_5 \end{bmatrix} = \begin{bmatrix} -3\text{ mA} \\ 12\text{ mA} \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

Step 5 of 6

The following is the MATLAB code for calculating loop currents.

```
a=[0 0 0 1 0;4 -3 -1 0 0;3 -7 0 4 0;1 0 -15 8 6;0 0 3 4 -8];b=[-0.003;0.012;0;0;0];c=inv(a)*b
```

Step 6 of 6

MATLAB output:

c =0.001619553666312-0.001020191285866-0.002461211477152-0.003000000000000-0.002422954303932

The values of the mesh currents are,

$$I_1 = 1.6195 \text{ mA}$$
$$I_2 = -1.0202 \text{ mA}$$
$$I_3 = -2.4612 \text{ mA}$$
$$I_4 = -3 \text{ mA}$$
$$I_5 = -2.423 \text{ mA}$$

Therefore, the values of the mesh currents, I_1, I_2, I_3, I_4, I_5 in the circuit shown in Figure 1 are $\boxed{1.6195 \text{ mA},\ -1.0202 \text{ mA},\ -2.4612 \text{ mA},\ -3 \text{ mA},\ -2.423 \text{ mA}}$ respectively.